Bayesian Graphical Lasso for Estimating Brain Networks

Liam Hanson, BIOS 731

Overview

- Discuss why we might want to estimate a graph
- Discuss the Bayesian Graphical Lasso Gibbs Sampler
- Review results from a small simulation study

Motivation

We are interested in viewing the brain as a graph i.e. a set of nodes, say regions of the brain, and edges, the connections between these edges

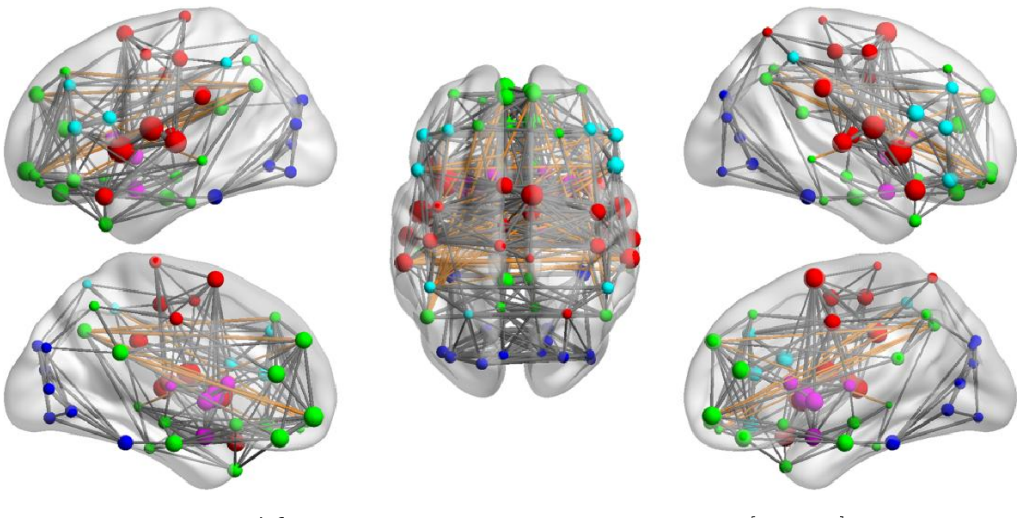


Figure 1: [@Hendricksen2015VisualizingDB]

Different brains have different network structures and therefore different local and global properties of these networks

Motivation

In general, healthy brains have a so-called "small-world" or "scale-free" structure.

Each node has few neighbors but it is easy for one group of nodes or sub-network to talk to another (six degrees of separation).

This implies the existence of hub nodes, nodes through which other areas of the network communicate.

Motivation

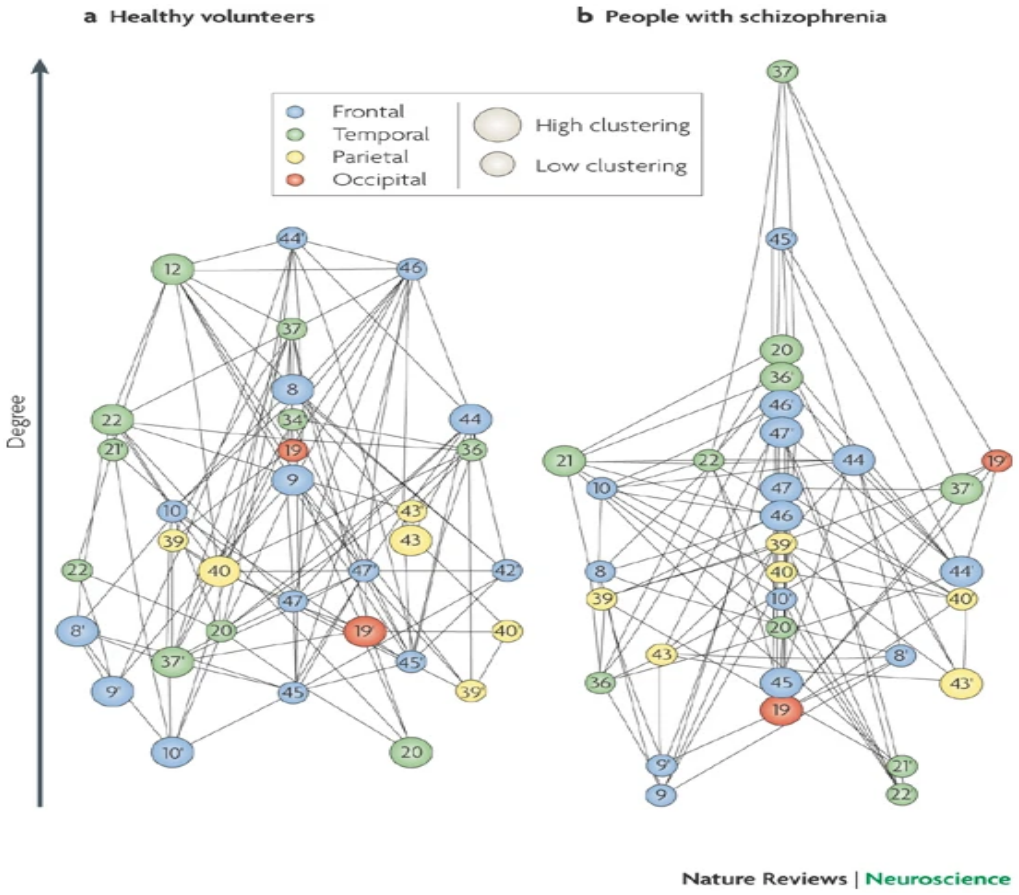


Figure 2: [@Bullmore2009]

A contrasting example is in people with Schizophrenia. These individuals tend to score lower on these small-world metrics.

Graph Estimation

We don't observe these graphs directly

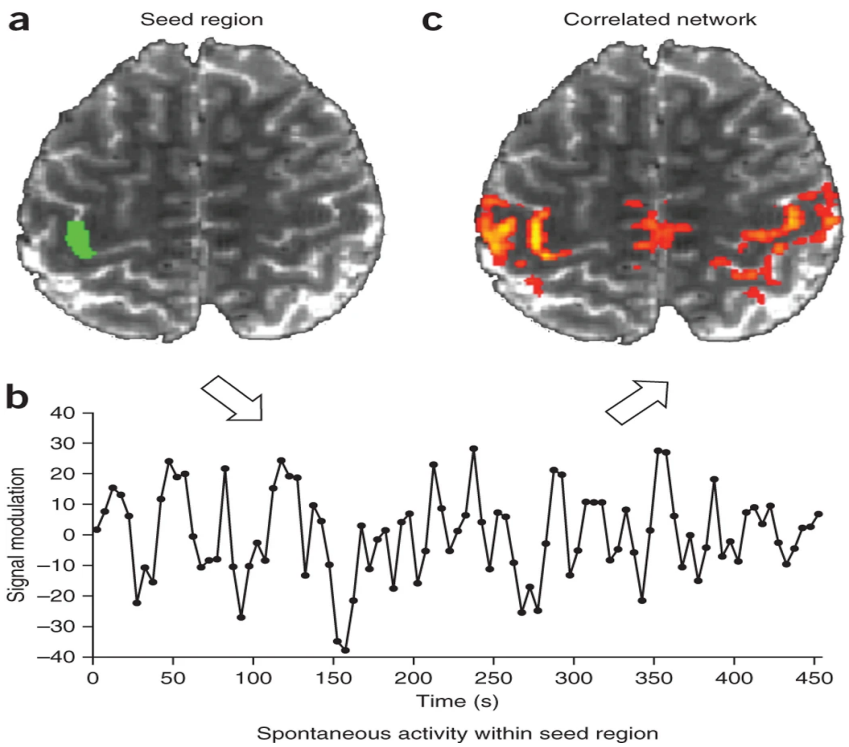


Figure 3: [@Buckner2013]

Instead, we have time series data for each voxel or region of interest in the brain where the measured quantity is the so-called BOLD signal.

Graph Estimation

A graph can be represented as an adjacency matrix: a symmetric matrix A with $a_{ij} = 1$ if an edge is present between node i, j and $a_{ij} = 0$ otherwise.

For our purposes, we assume that at time step (i.e. scan) t, the value of the BOLD signal across all D nodes $Y_D \sim MVN(0, \Sigma)$.

Fact: The precision matrix $\Omega = \Sigma^{-1}$ has the property that its entries ω_{ij} encode the partial correlation between node i and node j.

If $\omega_{ij} = 0$, these nodes are conditionally independent. Generally, we expect the true precision matrices to be sparse (recall scale free organization).

Graphical Lasso

- To capture this sparsity, something like LASSO (Tibshirani 1996) could work.
- We could fit successive pairwise LASSO's and regress the residuals to obtain partial correlations, but this isn't ideal
- Instead, let's maximize the penalized Multivariate Gaussian log-likelihood with respect to the Ω:

$$\log\,(\det\,\Omega) - \mathrm{tr}(\frac{S}{n}\Omega) - \rho||\Omega||_1$$

where S is the sample covariance matrix and $||\Omega||_1$ is the entry-wise l_1-norm of Ω.

Graphical Lasso

- This is the approach of Friedman, Hastie, and Tibshirani (2007)
- At a high level, it recursively fits a modified LASSO in a blockwise fashion using sub-gradient methods and coordinate descent (gets around non-differentiability of l_1 norm.)
- The focus of my project was a Bayesian Gibbs-Sampler approach to the Graphical Lasso by Wang (2012) to estimate the aforementioned graphs

Bayesian Graphical Lasso via Gibbs Sampling

We have the likelihood and prior

$$p(\mathbf{y}_t|\Omega) = MVN(\mathbf{y}_t|0, \Omega^{-1})$$

$$p(\Omega|\lambda) = C^{-1} \prod_{i<j} \mathrm{DE}(\omega_{ij}|\lambda) \prod_{i=1}^{p} \mathrm{Exp}(\omega_{ii}|\frac{\lambda}{2}) \cdot 1_{\Omega \in M^+}$$

where M^+ is the space of positive-definite matrices.

- DE is the double-exponential (Laplace) distribution used in classic Bayesian LASSO which induces potential shrinkage to 0.
- Natural exponential prior on the diagonal elements i.e. inverted variances.

Bayesian Graphical Lasso

- Fact: The double exponential results from marginalizing over a mixture of normals with variances τ_{ij} and exponential mixing density.
- This results in the posterior distribution:

$$p(\Omega, \tau | \mathbf{Y}, \lambda) \propto |\Omega|^{\frac{n}{2}} \exp\{\text{-tr } (\frac{1}{2} S\Omega)\} \prod_{i<j} \{\tau_{ij}^{-\frac{1}{2}} \exp(-\frac{\omega_{ij}^2}{2\tau_{ij}}) \exp(-\frac{\lambda^2}{2}\tau_{ij})\}$$

$$\times \prod_{i=1}^{p} \exp(\frac{\lambda}{2}\omega_{ij}) \cdot 1_{\Omega \in M^+}$$

- The restriction to the space of positive definite matrices makes this tricky, but we can work with it.

Deriving Gibbs Updates

For each update, partition Ω, S, and $\mathcal{T} = \{\tau_{ij}\}_{\in D \times D}$

$$\Omega = \begin{bmatrix} \Omega_{11} & \boldsymbol{\omega}_{12} \\ \boldsymbol{\omega}'_{12} & \omega_{22} \end{bmatrix} S = \begin{bmatrix} S_{11} & \boldsymbol{s}_{12} \\ \boldsymbol{s}'_{12} & s_{22} \end{bmatrix} \mathcal{T} = \begin{bmatrix} \mathcal{T}_{11} & \boldsymbol{\tau}_{12} \\ \boldsymbol{\tau}'_{12} & 0 \end{bmatrix}$$

where $\Omega_{11}, S_{11}, \mathcal{T}_{11}$ are block matrices and $\boldsymbol{\omega}_{12}, \boldsymbol{s}_{12}, \boldsymbol{\tau}_{12}$ are length $D-1$ vectors containing the joint precisions, sample covariances, and scaling factors for the current update.

Deriving Gibbs Updates

Then the conditional joint distribution of $(\boldsymbol{\omega}_{12}, \omega_{22})$ is

$$p(\boldsymbol{\omega}_{12}, \omega_{22} | \Omega_{11}, \mathcal{T}, \mathbf{Y}, \lambda) \propto (\omega_{22} - \boldsymbol{\omega}_{12}' \Omega_{11}^{-1} \boldsymbol{\omega}_{12})^{\frac{n}{2}}$$

$$\times \exp[-\frac{1}{2}\{\boldsymbol{\omega}_{12}' D_{\tau}^{-1} \boldsymbol{\omega}_{12} + 2 s_{12}' \boldsymbol{\omega}_{12} + (s_{22} + \lambda)\omega_{22}\}]$$

by dropping out constant terms and using determinant properties.

We then make the change of variables

$$(\boldsymbol{\omega}_{12}, \omega_{22}) \rightarrow (\boldsymbol{\beta} = \boldsymbol{\omega}_{12}, \gamma = \omega_{22} - \boldsymbol{\omega}_{12}' \Omega_{11}^{-1} \boldsymbol{\omega}_{12})$$

resulting in new joint conditional . . .

Deriving Gibbs Updates

$$(\gamma, \boldsymbol{\beta})|(\Omega_{11}, \mathcal{T}, \mathbf{Y}, \lambda) \sim \mathrm{Gamma}(\frac{n}{2} + 1, \frac{s_{22} + \lambda}{2}) \times \mathrm{N}(-\mathbf{C}s_{21}, \mathbf{C})$$

where $\mathbf{C} = \{(s_{22} + \lambda)\Omega_{11}^{-1} + D_{\tau}^{-1}\}^{-1}$

To sample from the full conditional of τ, use:

Fact: Conditional posteriors of $u_{ij} = \frac{1}{\tau_{ij}}$ are independently inverse Gaussian with parameters $\mu' = \sqrt{\lambda^2/\omega_{ij}^2}, \lambda' = \lambda^2$.

Finally, we sample from the full conditional of our tuning parameter λ given a prior $\lambda \sim \mathrm{Gamma}(r, s)$:

$$\lambda \sim \mathrm{Gamma}(r + D(D+1)/2, s + ||\Omega||_1/2)$$

Final Algorithm

This leads to the algorithm for the Block Gibbs Sampler. Given current Ω and τ:

1. For $i = 1, ..., D$

(a) Partition Ω, S and $\mathcal{T}$ as shown
(b) Sample $\gamma \sim (\Omega_{11}, \mathcal{T}, \mathbf{Y}, \lambda) \sim \mathrm{Gamma}(\frac{n}{2} + 1, \frac{s_{22}+\lambda}{2})$ and $\beta \sim \mathrm{N}(-\mathbf{C}s_{21}, \mathbf{C})$
(c) Update $\boldsymbol{\omega}_{12} = \beta, \omega_{22} = \gamma + \beta' \Omega_{11}^{-1} \beta$

2. Sample λ from $\lambda \sim \mathrm{Gamma}(r + D(D+1)/2, s + ||\Omega||_1/2)$
3. For $i < j$, sample $u_{ij} \sim \mathrm{Inv\text{-}Gau}(\mu', \lambda')$ and update $\tau_{ij} = \frac{1}{u_{ij}}$

Implementation and Simulation Study

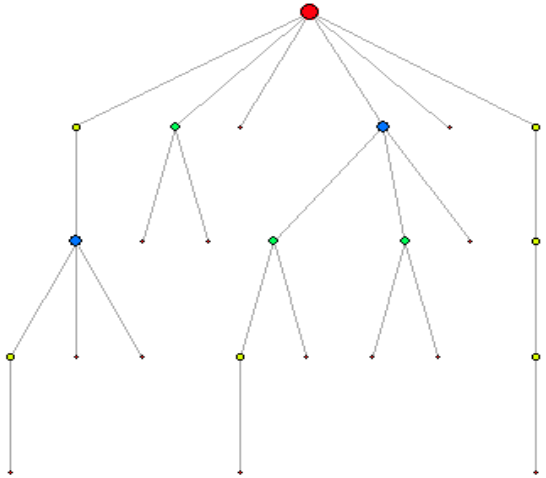
- We will compare the performance of the Bayesian Graphical Lasso to its frequentist counterpart with a focus on edge classification
- For Graphical Lasso, we simply say an edge exists between node i and node j if $\omega_{ij} \neq 0$. Any edges which shrunk to 0 are said to not exist.
- For the Bayesian Graphical Lasso, the probability of our posterior estimates being exactly 0 is 0, so we say an edge exists if a 95% credible interval about the posterior mean of ω_{ij} does not contain 0.
- Finally, for each simulation, we train the Graphical Lasso with 10-fold cross validation and use its estimates of Ω and λ as our initial Gibbs values (cheating).

Data

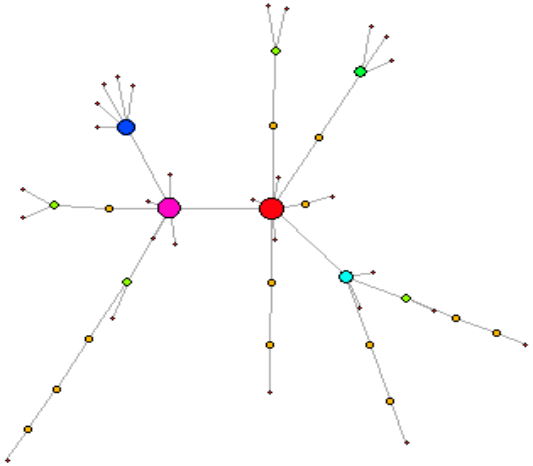
- We have four simulation scenarios with $D \in 25, 50, 100, 150$ and max timepoints $T = 200$.
- To simulate functional MRI data, we generate scale-free networks with D nodes then randomly rewire each node them with a small probability to mimic real brain networks.
- That graph is converted into an adjacency matrix and then each non-zero edge is randomly scaled to create a sparse precision matrix $\Omega_{true} \in D \times D$
- We then sample T times from $MVN(0, \Omega_{true})$ to generate our simulated functional data.

Simulation Results

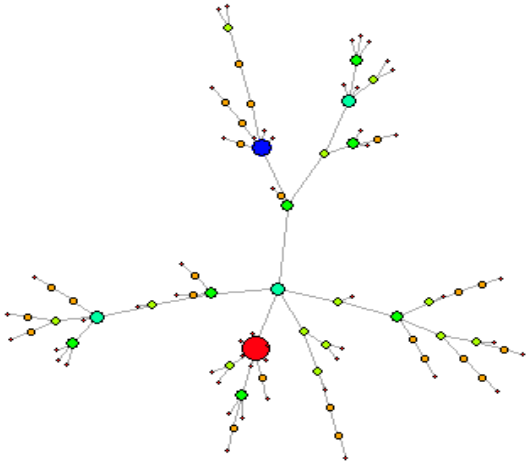
D = 25



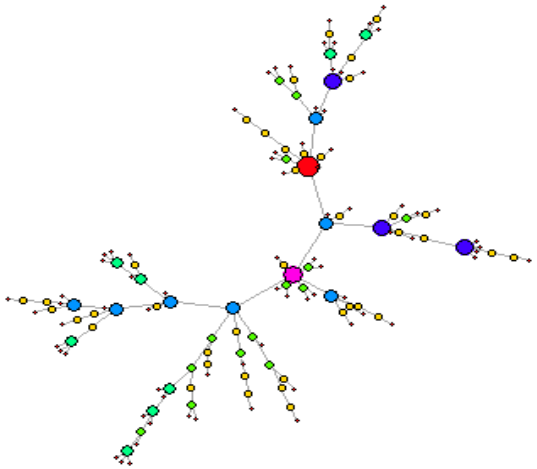
D = 50



D = 100

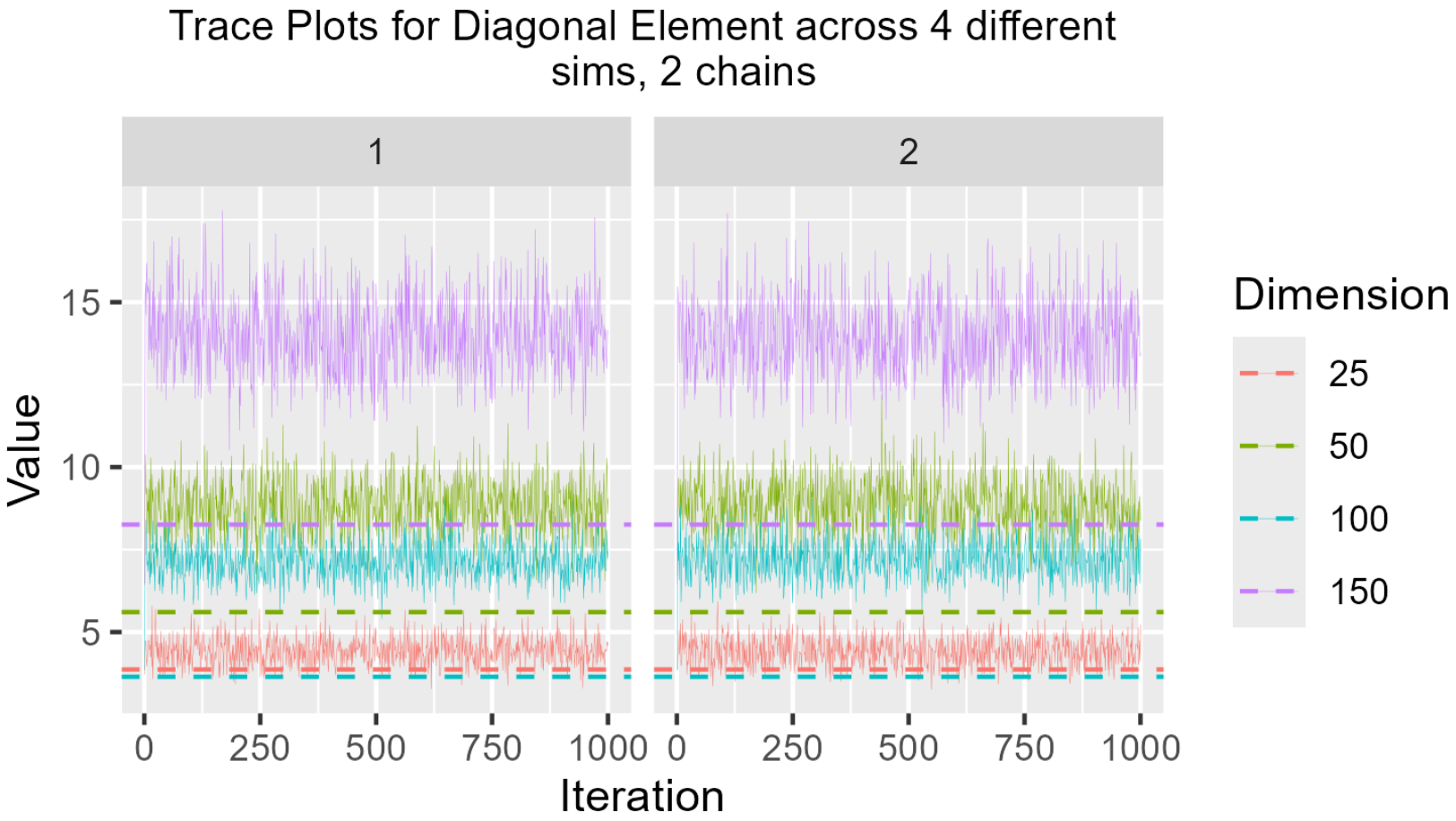


D = 150

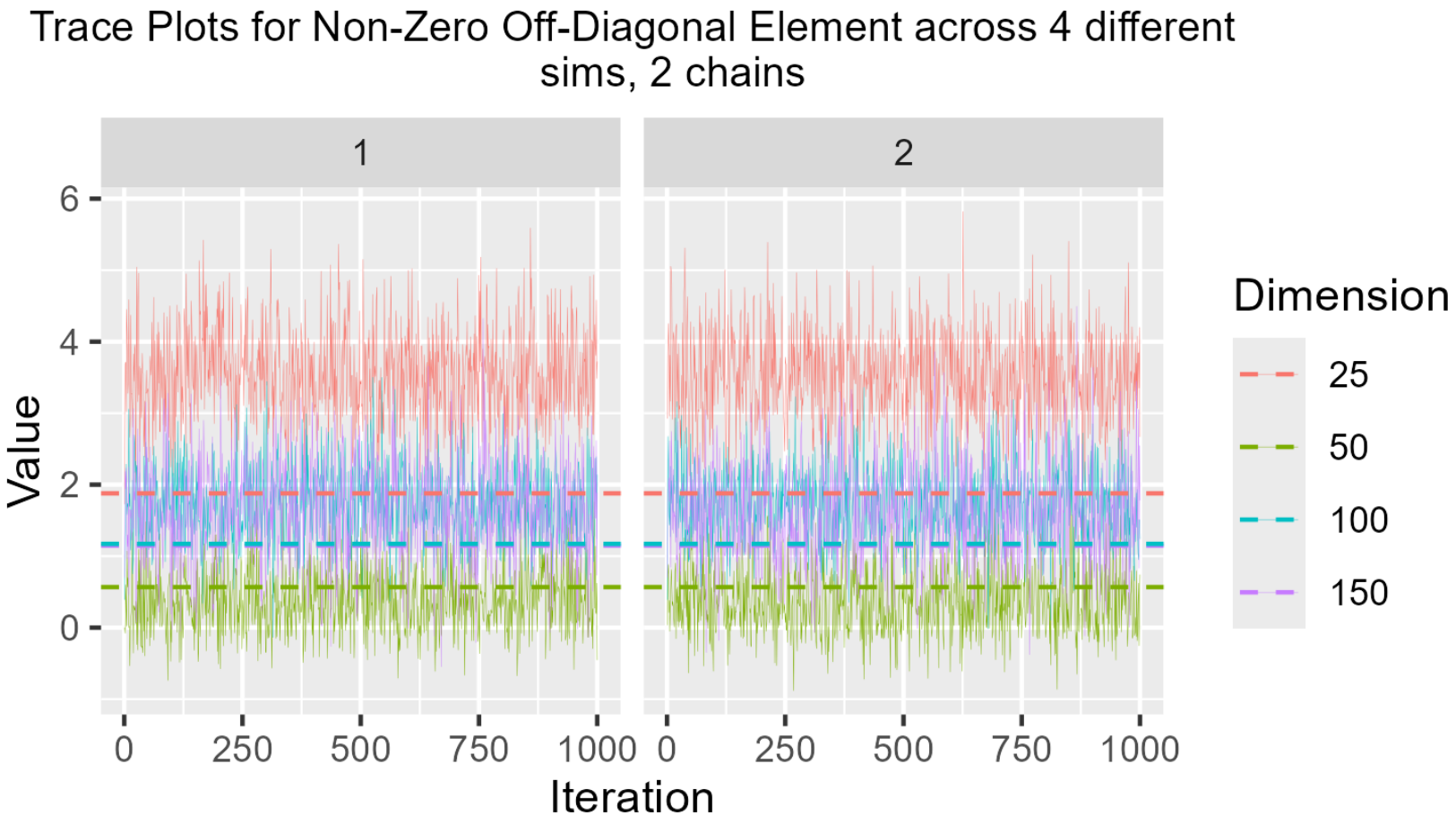


Simulation Results

Each simulation had 1000 samples with 200 burn-in samples. The sample appeared to mix well quickly.

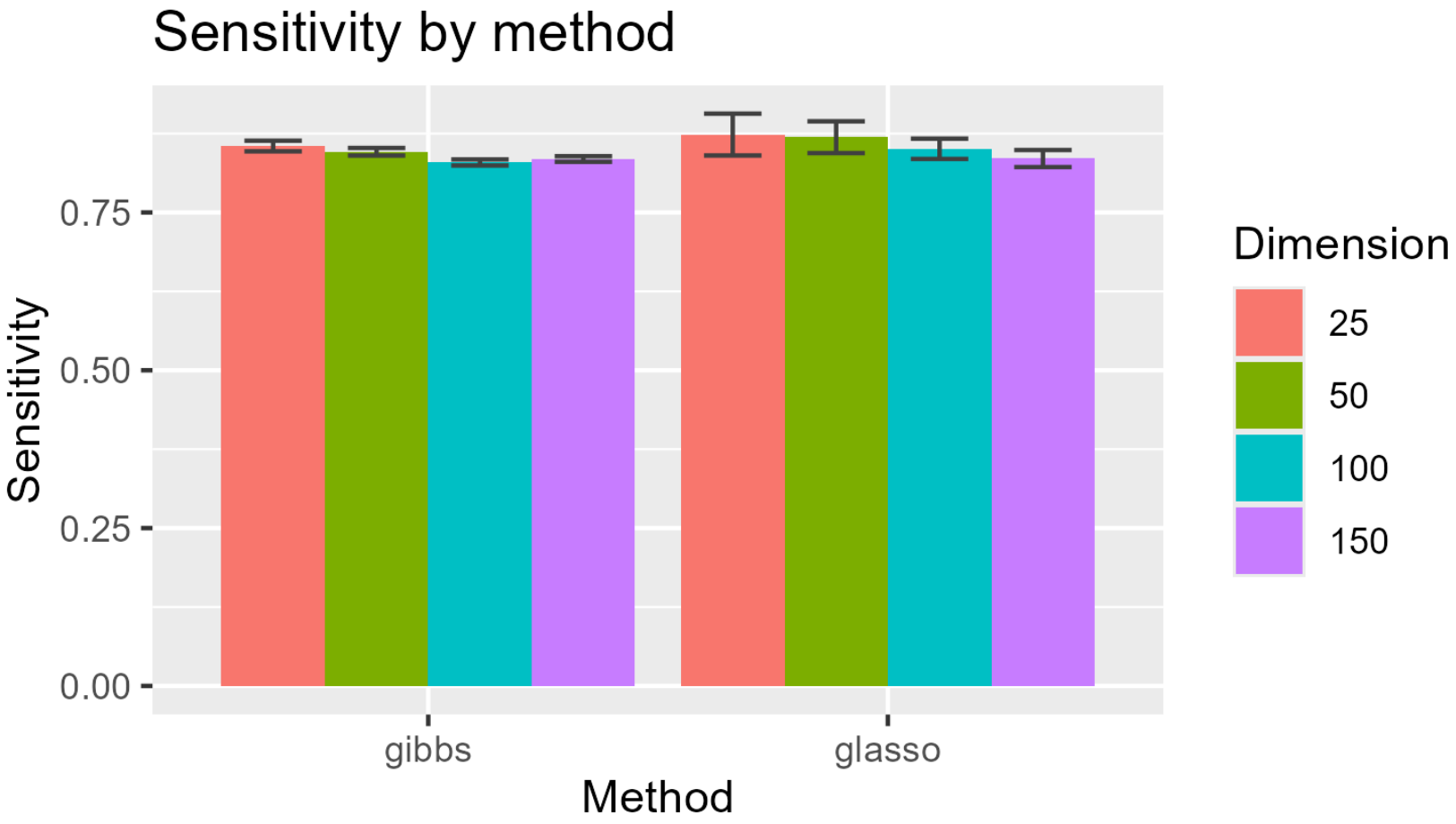


Simulation Results: Sampling

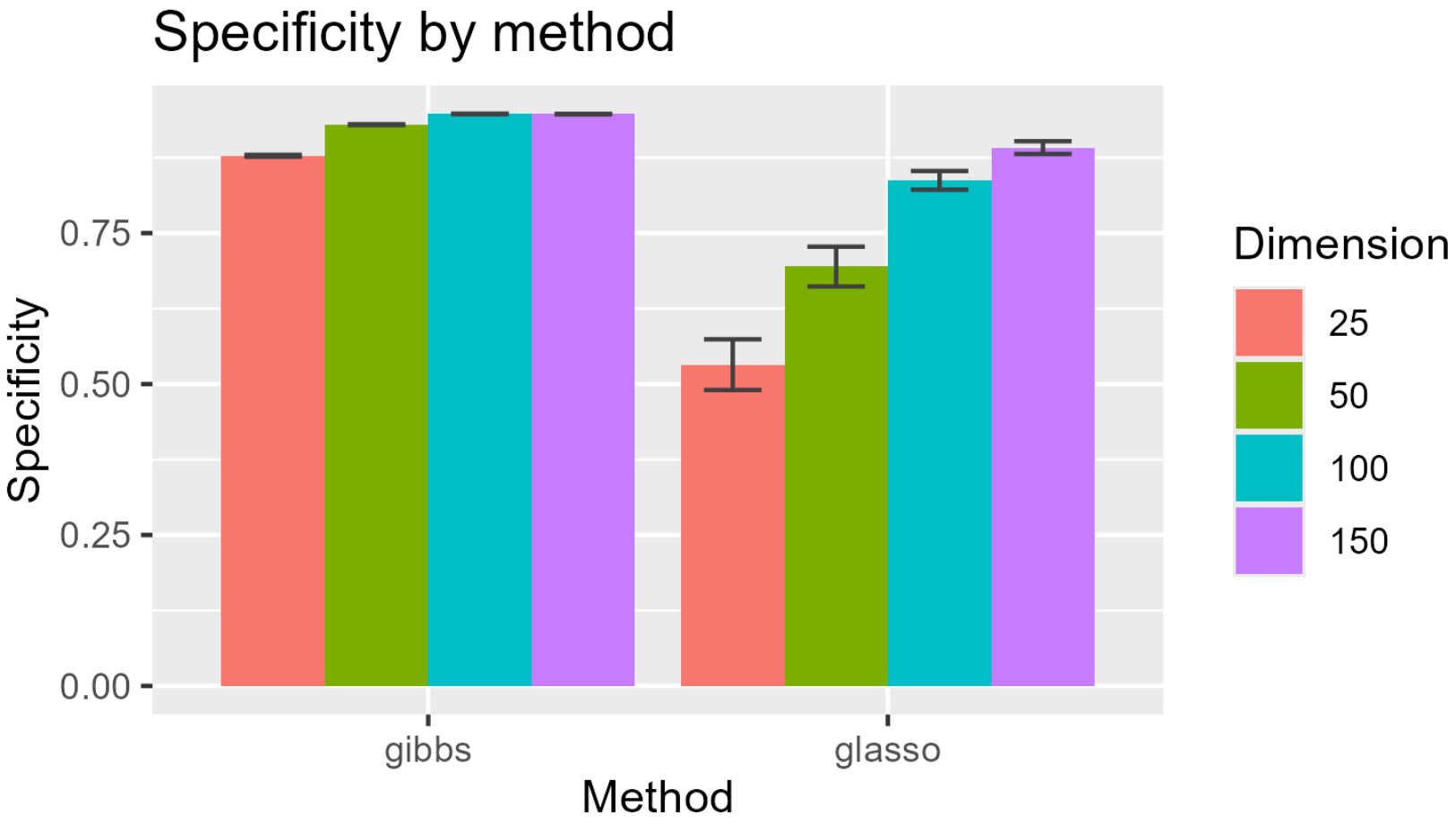


Simulation Results: Sampling

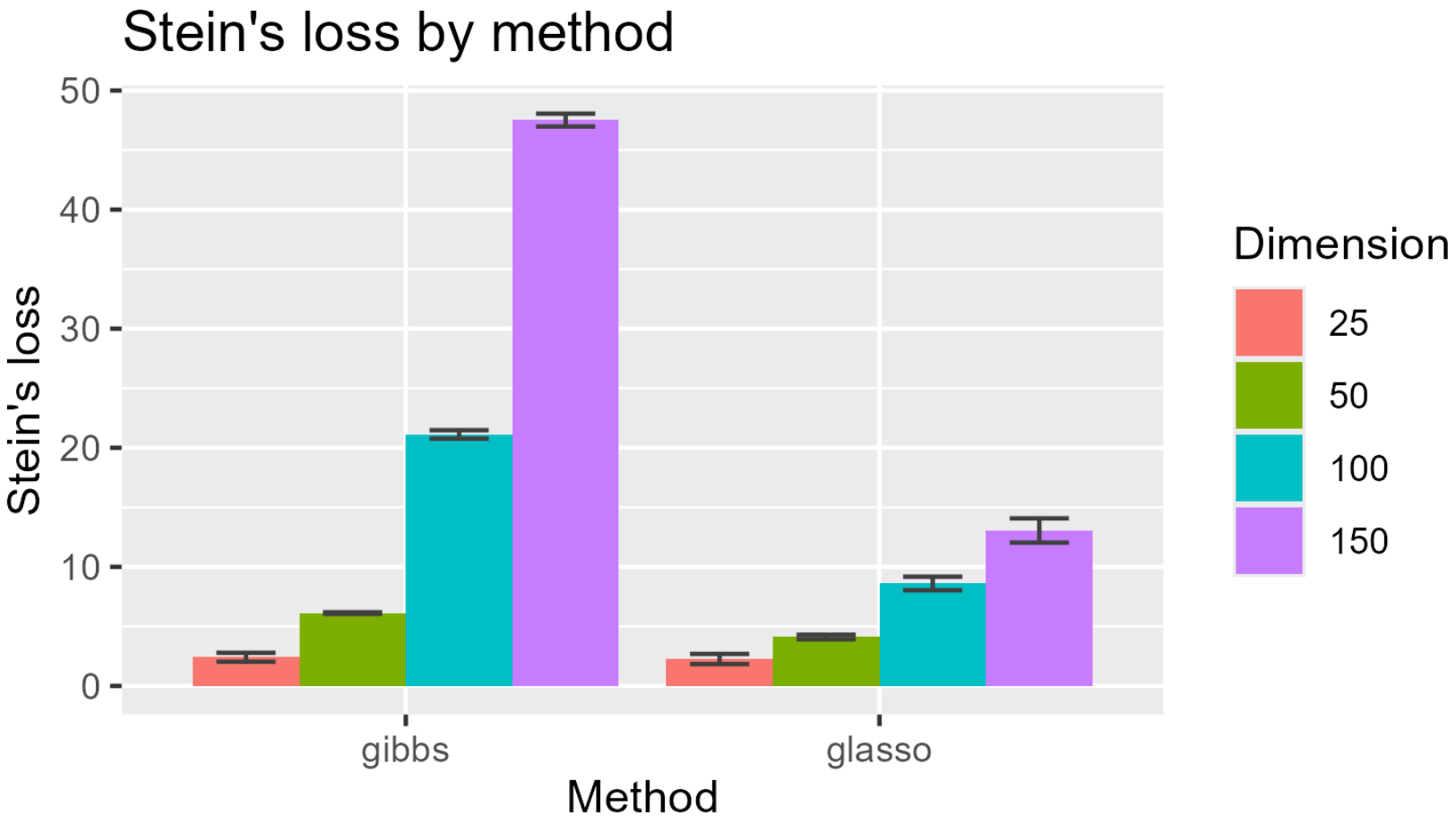
Simulation Results: Edge Classification



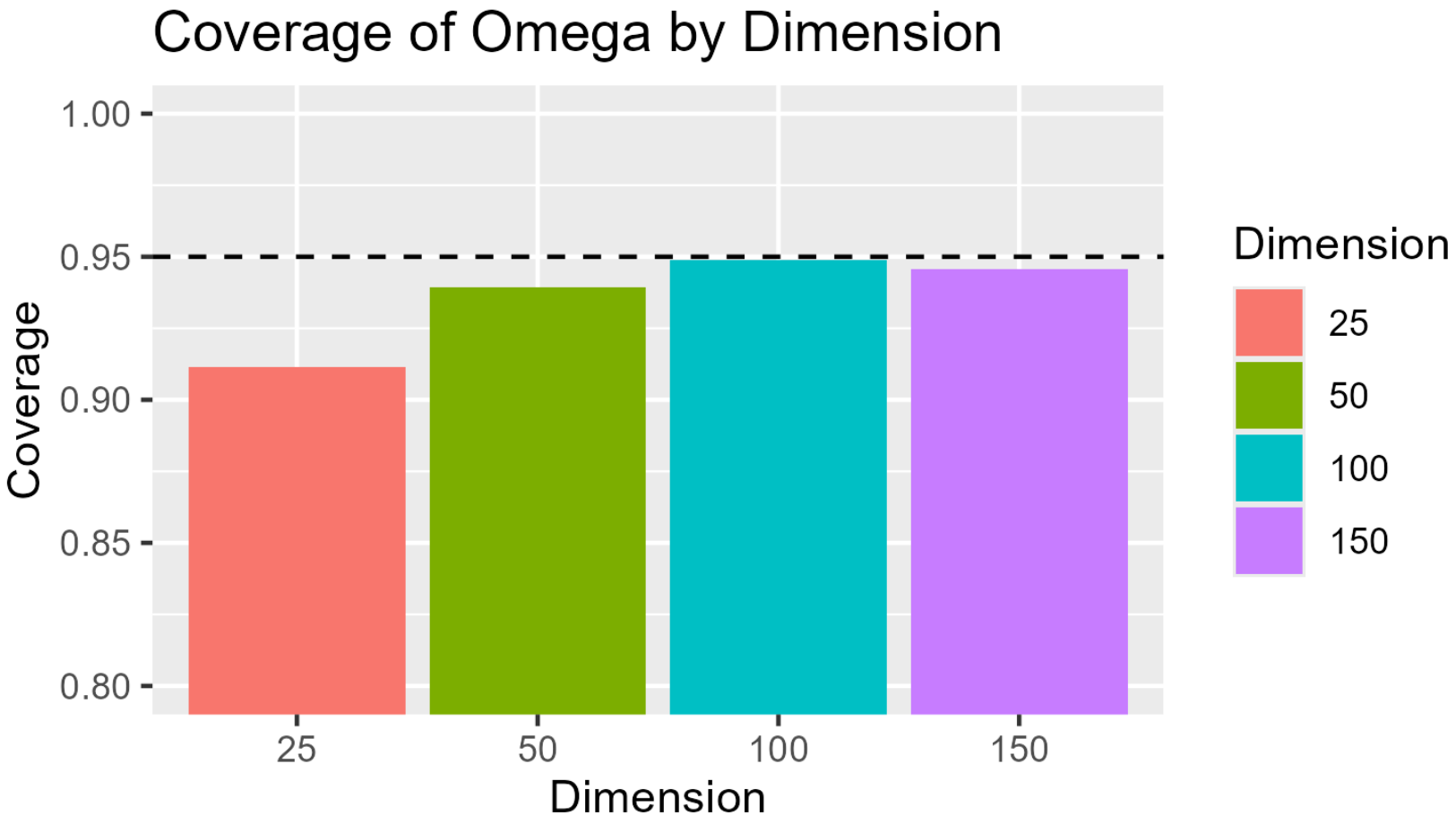
Simulation Results: Edge Classification



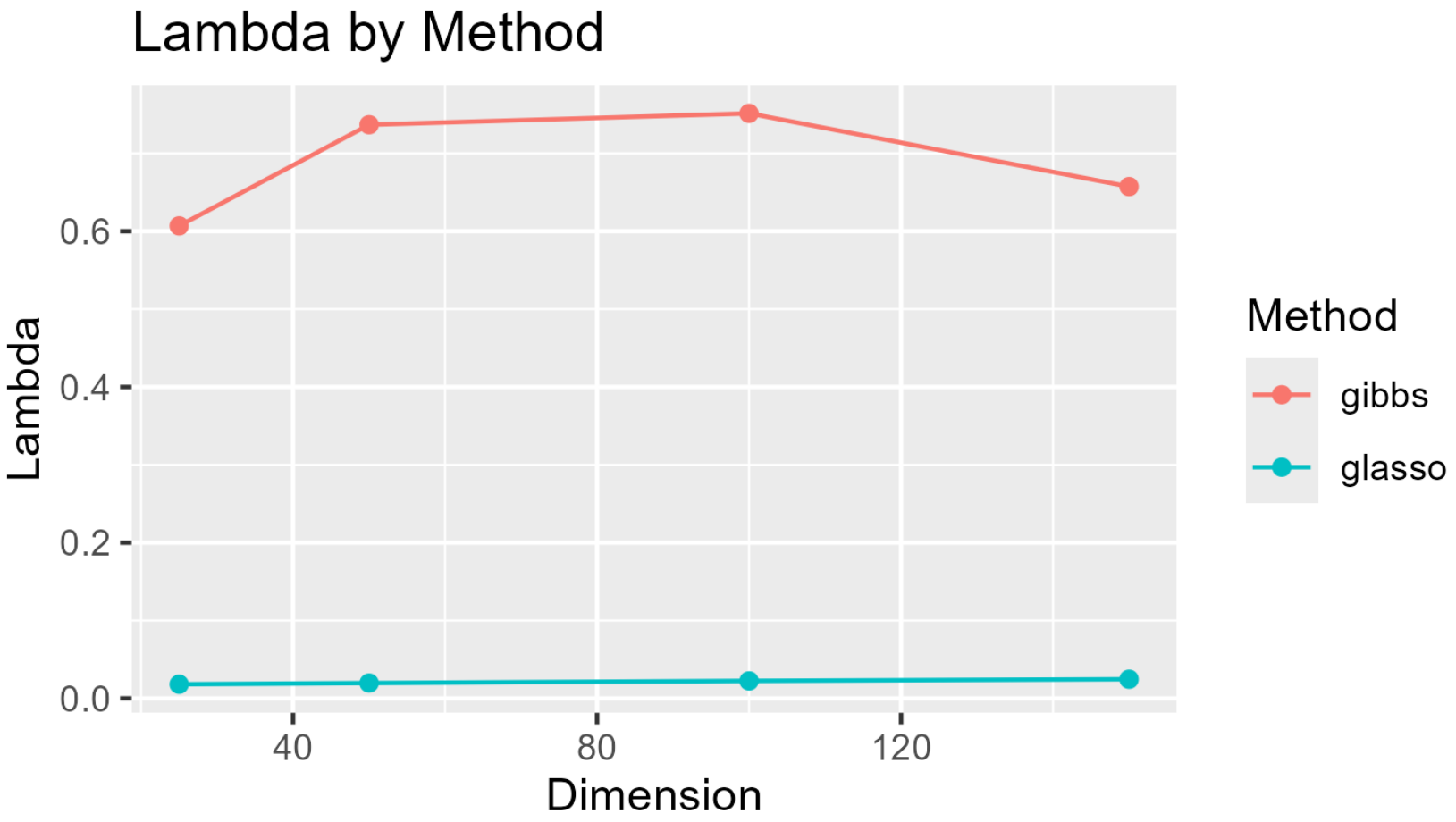
Simulation Results: Stein Loss



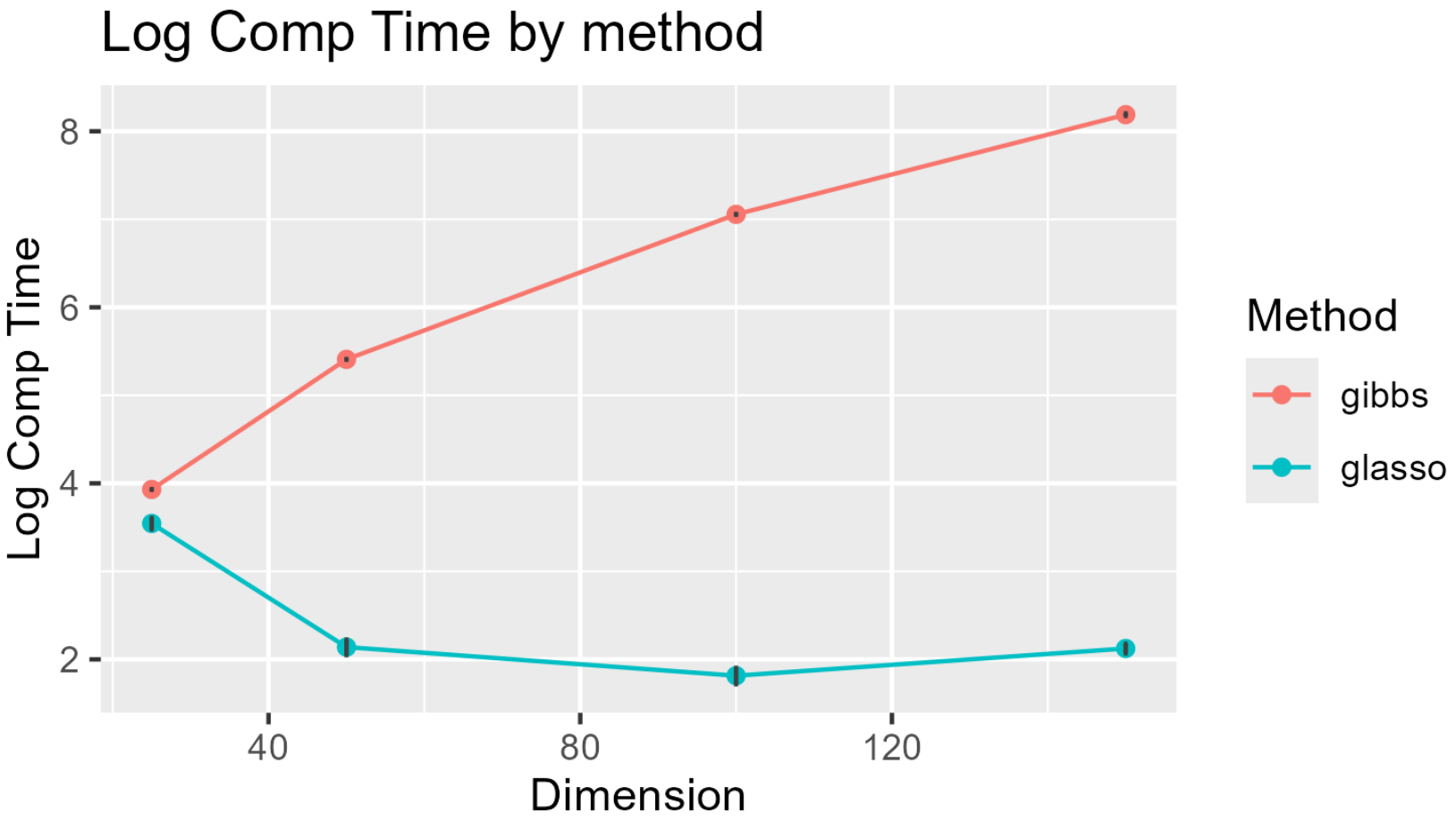
Simulation Results: Coverage



Simulation Results: Lambda



Simulation Results: Computation Time



Challenges

- Beastly computation time for D > 150. Brain data even at coarse parcellations has higher dimension.
- Extremely small values for τ matrices challenges invertibility

Future Directions

- Speed up computation dramatically, probably via Rcpp, so that this can be applied to real imaging data with higher dimension.
- Implement Bayesian Classification rule
- Bayesian Adaptive Graphical Lasso: allows for different λ's at different edges.

References

Friedman, Jerome, Trevor Hastie, and Robert Tibshirani. 2007. "Sparse Inverse Covariance Estimation with the Lasso." https://arxiv.org/abs/0708.3517.

Wang, Hao. 2012. "Bayesian Graphical Lasso Models and Efficient Posterior Computation." *Bayesian Analysis* 7 (4): 867–86. https://doi.org/10.1214/12-BA729.